NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83536/19
AMENDMENT 3
10 July 1999
SUPERSEDING
AMENDMENT 2
30 September 1996

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, TRANSIENT SUPPRESSED DC COIL

This amendment forms a part of MIL-PRF-83536/19, dated 27 March 1992, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

FIGURE 1, side view, dimension 1.025: Add "SQ".

PAGE 2

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

PAGE 3

FIGURE 1, side view: Add "MARKING (SEE NOTES 7 AND 8)" and draw arrow to relay.

PAGE 5

* NOTE 9, delete and substitute “9. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.”

* NOTE 10, delete and substitute “10. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 15 to 35, thickness .050 ±.005. Gasket material according to AMS 3332 has been considered acceptable.”

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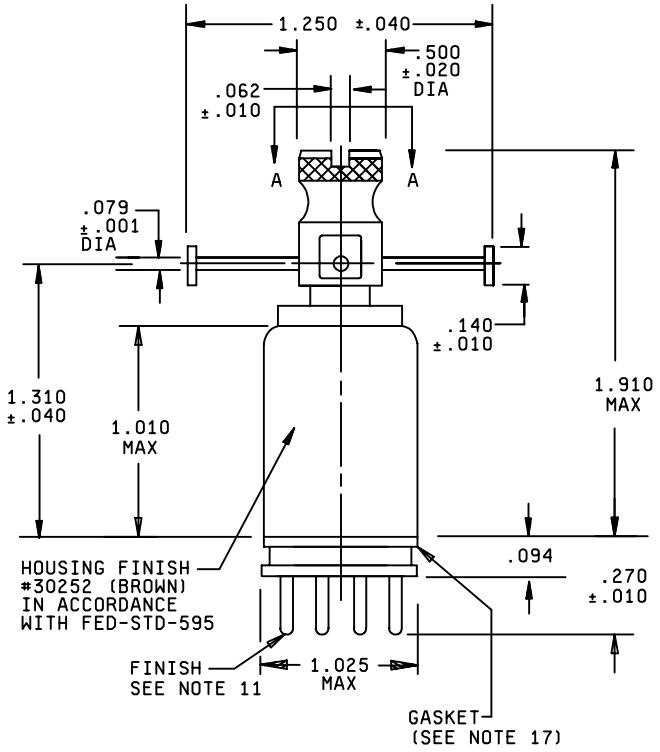
PAGE 6

* FIGURE 2, side view, delete and substitute the following:

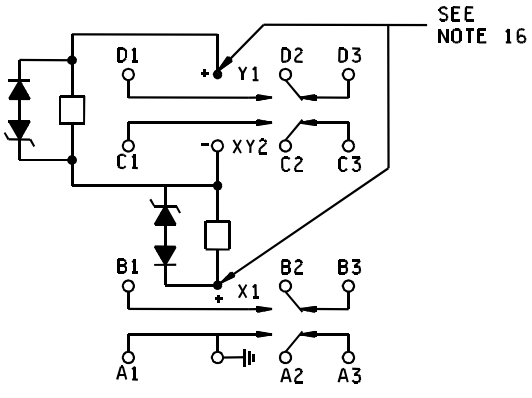
"



PAGE 7

FIGURE 2, add the following circuit diagram:

"



CIRCUIT DIAGRAM (TRACK MOUNT)
(ALTERNATE INTERNAL
DIODE CONFIGURATION)

"

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* NOTE 11, delete and substitute, “11. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.”

* NOTE 13, delete and substitute, “13. Track mount base, knob, latching pin and post shall be corrosion resistant. One way of ensuring corrosion resistance is to nickel plate stainless steel 303 with 100 microinches minimum thickness per McDonnell Douglas P.S.13112 (Deutsch Relays, Inc. Patent number 3.790,915) or equivalent 100 microinches thick minimum.”

* Add a NOTE 17, “17. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 30 to 45, thickness .050±.005. Gasket material according to AMS 3332 has been considered acceptable.”

PAGE 8

Delete "Contact voltage drop or resistance:" and substitute "Contact voltage drop and resistance:".

Break bounce requirement, delete in its entirety.

PAGE 9

Delete "Coil transient suppression: Applicable." and substitute "Coil transient suppression: Applicable (42 V dc maximum for track mount and 5 V dc maximum for all others)."

PAGE 10

TABLE III, delete and substitute:

"TABLE III. Dash numbers and characteristics. 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum pickup voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil current (amperes) maximum	Specified latch/ reset voltage (V dc) 4/	Specified latch/ reset voltage (V dc) 4/	High temper-ature test	Contin-uous curren t test
			Rated	Max					
001	002	RVFM	6	7.3	0.300	3.5	4.5	5.0	5.7
003		HFM	6	7.3	0.300	3.5	4.5	5.0	5.7
004	005	RVFM	12	14.5	0.150	6.5	9.0	9.9	11.25
006		HFM	12	14.5	0.150	6.5	9.0	9.9	11.25
007	008	RVFM	28	29 5/	0.072	14.5	8.0	19.8	22.5
009		HFM	28	29 5/	0.072	14.5	18.0	19.8	22.5
	010	Track	28	29 5/	0.072	14.5	18.0	19.8	22.5
011	012	RVFM	48	50	0.038	24.0	36.0	38.0	42.0
013		HFM	48	50	0.038	24.0	36.0	38.0	42.0

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1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc."

PAGE 11

Dielectric withstanding voltage, Track mount, All other points, Initial: Delete "1000" and substitute "1500".

Dielectric withstanding voltage, Track mount, Coil to coil/Coil to case, Sea level, After life: Delete "1250" and substitute "1050".

Dielectric withstanding voltage, Track mount, Altitude, At 300,000 feet: Delete "500" and substitute "N/A" (2 places).

Maximum leakage current, delete in its entirety and substitute: "Maximum leakage current for track mount only: 100 microamperes rms."

Delete "Shock" and all "Vibration" paragraphs and substitute:

"Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

"Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz. Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

"Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202, method 214, test condition IG (0.4 g^2/Hz, 50 to 2000 Hz); for horizontal flange mount, test condition IE (0.2 g^2/Hz) applies. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

"Vibration (random): Applicable to track mount only (qualification only). Test in accordance with MIL-STD-202, method 214, test condition IG, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts."

Weight: Delete "(77.2 g's)" and "(90.9 g's)" and substitute "(77.2 grams)" and "(90.9 grams)".

PAGE 13

TABLE V, second column: Delete "028" and substitute "010".

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based upon the entire content, irrespective of the marginal notations and relationship to the last previous amendment.

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